

Worksheet 11: Confidence Intervals

Confidence Intervals

A confidence interval provides a range of plausible values for an unknown parameter based on observed data. We illustrate the construction using a classical example: estimating the mean of a normal distribution.

Example: Mean of a Normal Distribution

Suppose we observe i.i.d. data

$$X_1, X_2, \dots, X_n \sim \mathcal{N}(\mu, \sigma^2),$$

where μ is unknown and σ^2 is known. A natural estimator of μ is the sample mean:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i.$$

Then we know that

$$\bar{X} \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right) \Rightarrow Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim \mathcal{N}(0, 1)$$

Picking quantiles of the standard Gaussian

$$\Rightarrow \mathbb{P}\left(-z_{1-\alpha/2} \leq Z \leq z_{1-\alpha/2}\right) = 1 - \alpha.$$

Thus, a $(1 - \alpha)$ confidence interval for μ is:

$$\left[\bar{X} - z_{1-\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{1-\alpha/2} \frac{\sigma}{\sqrt{n}} \right].$$

When the Exact Distribution is Unknown

In many practical settings, the exact distribution of the estimator is unknown. In such cases, we often rely on **asymptotic approximations**. A key result is the Central Limit Theorem (CLT), which states that for i.i.d. data with finite variance,

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} \mathcal{N}(0, 1) \quad \text{as } n \rightarrow \infty.$$

This allows us to construct *approximate* confidence intervals even when the data are not normally distributed, by treating the standardized statistic as approximately normal for large n .

When we use asymptotic results like the CLT, the resulting confidence intervals are **approximate**. This has an important implication:

- The true coverage probability is **not exactly** $1 - \alpha$.
- Instead, the coverage approaches $1 - \alpha$ as the sample size n increases.

Exercises

In each of the following, I state the population (F), the parameter, the sample size, and the estimator. Set up a simulation to verify the coverage probability of the confidence interval.

1. $F = N(\mu, 1)$, $n = 100$, and $\hat{\mu} = n^{-1} \sum_{i=1}^n X_i$.
2. $F = N(\mu, \sigma^2)$, $n = 100$, and $\hat{\mu} = n^{-1} \sum_{i=1}^n X_i$ and $\hat{\sigma}^2$ is the sample variance. The exact distribution of pivot statistic is

$$\frac{\hat{\mu} - \mu}{\hat{\sigma}/\sqrt{n}} \sim t_{n-1}.$$

- i. Assess coverage of confidence interval using the above distribution.
- ii. Assess coverage of confidence interval using the fact that

$$\frac{\hat{\mu} - \mu}{\hat{\sigma}/\sqrt{n}} \xrightarrow{d} N(0, 1)$$

- iii. Repeat for $n = 10$ and $n = 1000$. What do you observe.
3. $F = \text{Bern}(p)$, estimator is proportion of successes $\hat{p} = n^{-1} \sum_{i=1}^n X_i$.
 - i. Consider the following approximation distribution and assess the coverage of a confidence interval based on this:

$$\sqrt{n}(\hat{p} - p) \xrightarrow{d} N(0, p(1 - p))$$

- ii. Consider the following approximation distribution and assess the coverage of a confidence interval based on this:

$$\frac{\sqrt{n}(\hat{p} - p)}{\hat{p}(1 - \hat{p})} \xrightarrow{d} N(0, 1)$$

- iii. Compare the coverages for different n .
4. $F = \text{Gamma}(\alpha, 1)$, $n = 50$, and estimator is sample mean for $\alpha = .01$.
 5. Consider $F = N_2(\mu, \text{diag}(1, 1))$, $n = 100$, and estimator of μ is the sample mean. Let $\mu = (\mu_1, \mu_2)$.
 - i. Calculate the coverage for μ_1 .
 - ii. Calculate the coverage for μ_2 .
 - iii. Calculate the coverage for $\mu = (\mu_1, \mu_2)$. What do you notice?